

$$\begin{aligned}
 1, \quad G_{nf}(s) &= \frac{s^2 + 2\zeta_z \omega_n s + \omega_n^2}{s^2 + 2\zeta_p \omega_n s + \omega_n^2} \\
 &= \frac{s^2 + 2\zeta_p \cdot d \cdot \omega_n s + \omega_n^2}{s^2 + 2\zeta_p \omega_n s + \omega_n^2} \\
 &\quad \text{令 } 2\zeta_p = \frac{1}{Q} \\
 &= \frac{s^2 + \frac{\omega_n}{Q} d s + \omega_n^2}{s^2 + \frac{\omega_n}{Q} s + \omega_n^2}
 \end{aligned}$$

★ $\begin{cases} Q = \text{Pt40A (Unit: 0.01)} \Rightarrow \text{越大, 寬度越窄, 相位損失越少} \\ d = \text{Pt40B (Unit: 0.001)} \Rightarrow \text{大} \end{cases}$ 少; $d=1$ 時, $G_{nf}(s)=1$

$$\begin{aligned}
 2, \quad s &= j\omega \text{ 代入} \\
 &\Rightarrow \frac{-\omega^2 + \frac{\omega_n}{Q} d \cdot j\omega + \omega_n^2}{-\omega^2 + \frac{\omega_n}{Q} \cdot j\omega + \omega_n^2} \\
 &= \frac{\omega_n^2 - \omega^2 + j \frac{1}{Q} d \cdot \omega_n \cdot \omega}{\omega_n^2 - \omega^2 + j \frac{1}{Q} \cdot \omega_n \cdot \omega}
 \end{aligned}$$

相位損失計算:

$$\text{phase_loss} = \text{atan}\left(\frac{\frac{1}{Q} d \cdot \omega_n \cdot \omega}{\omega_n^2 - \omega^2}\right) - \text{atan}\left(\frac{\frac{1}{Q} \cdot \omega_n \cdot \omega}{\omega_n^2 - \omega^2}\right)$$

$\omega_{\text{應}} < \omega_n \leftarrow \begin{cases} \omega_n = 2 \cdot \pi \cdot f_n \Rightarrow \text{Notch 壓的頻率} \\ \omega = 2 \cdot \pi \cdot f_{BW} \Rightarrow \text{要看的頻率 (可以帶速度頻寬)} \end{cases}$
(i.e. 不能 $\geq$)

ex. $f_n = 1480$, $f_{BW} = 300$; $\text{Pt40A} = 70$, $Q = 0.7$
 $\omega_n = 9299$, $\omega_{BW} = 1885$; $\text{Pt40B} = 100$, $d = 0.1$

(i.e. 當 Notch 壓在 1480 時, 在速度迴路頻寬為 300 時, 會損失多少相位)

$$\Rightarrow \text{phase_loss} = \text{atan}\left(\frac{\frac{1}{0.7} \times 0.1 \times 9299 \times 1885}{9299^2 - 1885^2}\right) -$$

$$\text{atan}\left(\frac{\frac{1}{0.7} \times 9299 \times 1885}{9299^2 - 1885^2}\right)$$

$$= 1.73 - 16.8 = -15.07 \text{ 度}$$

(可以用 MATLAB 驗證)

* 可以省略 2π
因為約掉

$$\begin{aligned}
 \text{ex. } &\frac{\frac{1}{0.7} \times 2\pi \cdot f_n \times 2\pi \cdot f_{BW}}{(2\pi \cdot f_n)^2 - (2\pi \cdot f_{BW})^2} \\
 &= \frac{\frac{1}{0.7} \times 2\pi \cdot f_n \times 2\pi \cdot f_{BW}}{(2\pi)^2 \cdot f_n^2 - (2\pi)^2 \cdot f_{BW}^2} \\
 &= \frac{\frac{1}{0.7} \times \cancel{2\pi} \cdot f_n \times \cancel{2\pi} \cdot f_{BW}}{(\cancel{2\pi})^2 \cdot (f_n^2 - f_{BW}^2)}
 \end{aligned}$$

3. 所以, 壓Notch後, 可以先估算預期要搜尋的速度迴路頻寬上限, 其損失的相位.

故, 損失相位 $< TH$ (ex. 5度) $\Rightarrow$ 不用下修增益

$> TH$ $\Rightarrow$ 需下修增益, 用以下公式,

$$\text{phase_loss} = \text{atan}\left(\frac{\frac{1}{Q}d \cdot \omega_n \cdot \omega}{\omega_n^2 - \omega^2}\right) - \text{atan}\left(\frac{\frac{1}{Q} \cdot \omega_n \cdot \omega}{\omega_n^2 - \omega^2}\right)$$

$$\Rightarrow \begin{cases} \text{phase_loss} = TH = 5^\circ \\ \omega_n = \text{Notch壓的頻率} \\ \omega = \text{從原本預期的上限值, 往下遞減, 直到} \leq TH, \text{就是下修後的頻寬} \end{cases}$$